Tutorial 1

Risk, Prospect Theory, and Bayesian updating

with step-by-step solutions

Marco Rosso

CLEF – University of Bologna

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How this tutorial works

- Five blocks. Each one opens with a compact recap of the theory and a pointer to the lecture where it is developed, and then moves to exercises on that same material.
- The exercises use the same templates as the online quizzes on Virtuale and as the mid-term. Same mechanics, different numbers.
- Try each one first. We solve it together afterwards.
- Questions on the *content* of the course go to Prof. Bigoni. Questions on the organisation, the quizzes and the exam come to me: `marco.rosso4@unibo.it`

Coverage: Lectures 1a, 1b, 2, 3a, 3b, 4a, 4b.

Plan

1. Experimental method: treatments, designs, control
2. Expected utility, risk attitudes, the Holt–Laury menu
3. The Allais paradox and Prospect Theory
4. Bayes' rule, base rates, representativeness
5. Probability matching

Sample *oral* questions at the end.

PART 1

Experimental method

Experimental method

Lecture 1a – "Treatment", "Between- and within-subject designs", "Advantages of experiments"

Treatment = fully specified set of procedures: instructions, incentives, rules of play. Vary **one thing at a time**.

	each subject sees	the threat to identification
Between-subject	one treatment	composition of the two groups
Within-subject	two or more	order / sequence effects

Control: clear incentives, unbiased terminology, no deception.

Confound: T1 = small + real, T2 = large + hypothetical $\Rightarrow$ two factors move together, neither is identified.

Exercise 1.1

Design

Consider a design in which each participant takes part in one session only, and each session is run under a single set of instructions: half of the sessions use real payoffs, the other half hypothetical payoffs.

Is this a between-subject or a within-subject design?

a. Within-subject
b. Between-subject

Exercise 1.1 – step by step

Step 1. Ask: how many treatments does *each* participant see? Each person takes part in one session, and each session has a single set of instructions ⇒ one treatment per person.

Step 2. Ask: what is compared with what? Behaviour with real payoffs (one group) against behaviour with hypothetical payoffs (another group): two different groups of people.

Step 3. One treatment per subject, comparison across groups ⇒ **b. Between-subject**.

Step 4. Contrast: it would be within-subject if the *same* people made decisions under both real and hypothetical payoffs.

Trade-off to remember: between-subject avoids order effects; within-subject controls for individual differences.

Exercise 1.2

Control

In the lottery-choice experiment of Lecture 1b, all participants first made 10 decisions at the **1**× payoff scale, then the same 10 decisions at a high *hypothetical* scale, then at a high *real* scale, and finally again at **1**×. The number of safe choices rises with the real scale.

Why did Harrison et al. (2005) call this design a possible fatal flaw?

a. Because the subjects were students
b. Because payoff scale and order of presentation change together
c. Because the payoffs were too small
d. Because the design is between-subject

Exercise 1.2 – step by step

Step 1. List what changes across the menus: the payoff scale (**1**×, then high) *and* the position in the session (first, second, third, fourth).

Step 2. Check how the two are combined: the high real scale *always* comes after the **1**× and the hypothetical menus. Scale and order move together.

Step 3. Consequence: more safe choices at the high real scale could be due to the scale, or to having already made 20 decisions (learning, fatigue, order). The design cannot tell the two apart: control is lost ⇒ **b**.

Step 4. Discard the other options: a. and c. do not explain a confound; d. is false, every participant goes through all the menus (within-subject).

The fix: randomise the order of the menus across sessions.

PART 2

Expected utility and risk attitudes

Expected utility

Lecture 1b – "Expected payoff", "Expected utility", "Relative risk aversion"

Lottery L: x_1 with probability p, x_2 with probability $1-p$.

expected value	$\mathbb{E}[L] = p\,x_1 + (1-p)\,x_2$
expected utility	$\mathbb{E}[u(L)] = p\,u(x_1) + (1-p)\,u(x_2)$
certainty equivalent	$u(CE) = \mathbb{E}[u(L)] \;\Rightarrow\; CE = u^{-1}\big(\mathbb{E}[u(L)]\big)$
risk premium	$RP = \mathbb{E}[L] - CE$
CRRA	$u(x) = x^{1-r}$

u concave $\Leftrightarrow$ risk averse $\Leftrightarrow$ $CE < \mathbb{E}[L] \;\Leftrightarrow\; RP > 0 \;\Leftrightarrow\; 0 < r < 1$.

Switching point p^*: solves $\mathbb{E}[u(A)] = \mathbb{E}[u(B)]$. Switching later = more risk averse.

Exercise 2.1

A row of a lottery-choice menu

	payoff with prob. p	payoff with prob. $1-p$
Option A (safe)	€81	€25
Option B (risky)	€121	€1

Set $p = 0.5$.

(a) Compute the expected payoff of each option.
(b) The agent has $u(x) = \sqrt{x}$. Compute the expected utility of each option and the corresponding certainty equivalent. Which option is chosen?

Exercise 2.1 – step by step

(a) Step 1. Expected payoff $= p\,x_1 + (1-p)\,x_2$, with $p = 0.5$.

Step 2. A: $0.5 \times 81 + 0.5 \times 25 = 40.5 + 12.5 = 53$.

Step 3. B: $0.5 \times 121 + 0.5 \times 1 = 60.5 + 0.5 = 61$. B is higher by €8.

(b) Step 1. Utilities of the four payoffs: $\sqrt{81} = 9$, $\sqrt{25} = 5$, $\sqrt{121} = 11$, $\sqrt{1} = 1$.

Step 2. Expected utilities: A: $0.5 \times 9 + 0.5 \times 5 = 7$; B: $0.5 \times 11 + 0.5 \times 1 = 6$.

Step 3. Certainty equivalents, $u(CE) = \mathbb{E}[u]$, i.e. $\sqrt{CE} = \mathbb{E}[u]$, so $CE = \mathbb{E}[u]^2$: $CE_A = 7^2 = 49$, $CE_B = 6^2 = 36$.

Step 4. Compare expected utilities (or, equivalently, CEs): $7 > 6 \Rightarrow$ **A is chosen**.

Step 5. Risk premia, $\mathbb{E}[L] - CE$: A: $53 - 49 = 4$; B: $61 - 36 = 25$.

Typical mistake: comparing expected *payoffs* ($61 > 53$) and concluding B. The agent maximises expected *utility*.

Exercise 2.2

The switching point

Same two options as in Exercise 2.1, but now let p vary as you move down the rows of the menu.

(a) At which p does a *risk-neutral* agent switch from A to B?

(b) At which p does an agent with $u(x) = \sqrt{x}$ switch?

(c) What does the comparison tell you about how to read the data from a Holt–Laury menu?

Exercise 2.2 – step by step

(a) Step 1. Risk neutral: compare expected payoffs as functions of p.

$$\mathbb{E}[A] = 81p + 25(1-p) = 25 + 56p \qquad \mathbb{E}[B] = 121p + 1\,(1-p) = 1 + 120p$$

Step 2. Switching point: $25 + 56p = 1 + 120p \;\Rightarrow\; 24 = 64p \;\Rightarrow\; p = 24/64 = 3/8 = 0.375$.

(b) Step 1. Same with $u(x) = \sqrt{x}$:

$$\mathbb{E}[u(A)] = 9p + 5(1-p) = 5 + 4p \qquad \mathbb{E}[u(B)] = 11p + 1\,(1-p) = 1 + 10p$$

Step 2. $5 + 4p = 1 + 10p \;\Rightarrow\; 4 = 6p \;\Rightarrow\; p = 2/3 \approx 0.667$.

(c) Step 1. With rows $p = 0.1, 0.2, \ldots, 1$ as in Holt–Laury: the risk-neutral agent chooses A for $p \le 0.3$ (3 safe choices), the $\sqrt{\cdot}$ agent for $p \le 0.6$ (6 safe choices).

Step 2. The more risk averse, the later the switch: the number of safe choices measures risk aversion. In Holt and Laury's own menu risk neutrality implies four safe choices; subjects made about six.

Step 3. At $p = 0.5$ (Exercise 2.1) the risk-neutral agent has already switched to B, the $\sqrt{\cdot}$ agent has not.

Exercise 2.3

Certainty equivalent

A lottery pays €90 or €10, each with probability $1/2$. The agent has $u(x) = \sqrt{x}$.

Compute the expected value, the certainty equivalent and the risk premium.

How large is the risk premium as a share of the expected value?

Exercise 2.3 – step by step

Step 1. Expected value: $0.5 \times 90 + 0.5 \times 10 = 45 + 5 = 50$.

Step 2. Utilities: $\sqrt{90} = 9.4868$, $\sqrt{10} = 3.1623$.

Step 3. Expected utility: $0.5 \times 9.4868 + 0.5 \times 3.1623 = 4.7434 + 1.5811 = 6.3246$.

Step 4. Certainty equivalent: invert u, i.e. square: $CE = 6.3246^2 = 40.0$.

Step 5. Risk premium: $50 - 40 = 10$; as a share of the expected value: $10/50 = 20\%$.

Step 6. Optional check without rounding:

$$CE = \left(\frac{\sqrt{90} + \sqrt{10}}{2} \right)^2 = \frac{90 + 10 + 2\sqrt{90 \times 10}}{4} = \frac{100 + 2 \times 30}{4} = \frac{160}{4} = 40$$

Typical mistake: computing $\sqrt{50}$ (utility of the expected value) instead of the expected utility. $\sqrt{50} = 7.07 > 6.32$: that gap *is* risk aversion.

Exercise 2.4

Hypothetical bias

In Holt and Laury (2002) subjects also made the same ten choices knowing that they would *not* be paid for that part. Which statement is correct?

a. Subjects were more risk averse with hypothetical payoffs
b. Subjects were less risk averse with hypothetical payoffs
c. Behaviour was identical in the two conditions
d. Subjects made no mistakes with hypothetical payoffs

Exercise 2.4 – step by step

Step 1. Recall the measure: more safe choices in the menu = more risk aversion.

Step 2. Recall the data: about 6 safe choices with real payments, about 5 with hypothetical payments.

Step 3. Fewer safe choices with hypothetical payoffs ⇒ **b. less risk averse**.

Step 4. Discard the others: a. is the opposite; c. is contradicted by the 6 against 5; d. is false, since 2% chose a sure $40 over a sure (hypothetical) $77, a dominated choice.

Related: scaling up *real* payoffs raises the number of safe choices; scaling up *hypothetical* payoffs does not.

PART 3

The Allais paradox and Prospect Theory

The independence axiom

$$A \succsim B \qquad \Longleftrightarrow \qquad \alpha A + (1-\alpha) z \;\succsim\; \alpha B + (1-\alpha) z \qquad \alpha \in (0, 1]$$

With probability $1-\alpha$ you get z *whatever you chose*, with probability α you get the lottery you selected. z is the “irrelevant” common component.

Expected utility is **linear in probabilities**, so diluting both options by the same α rescales the comparison without reversing it:

$$u(x_B) > p\, u(x_A) \qquad \Longrightarrow \qquad \alpha\, u(x_B) > \alpha\, p\, u(x_A) \qquad \text{for every } u$$

The Allais paradox – the two problems

Lecture 2 – "The Allais paradox"; percentages are the modal responses

Problem 1			**Problem 2** = Problem 1 diluted by $\alpha = \frac{1}{4}$		
A:	4,000 with prob. 0.80	20%	A:	4,000 with prob. 0.20	**65%**
B:	3,000 with certainty	**80%**	B:	3,000 with prob. 0.25	35%

Both probabilities are multiplied by $\frac{1}{4}$: $0.80 \to 0.20$, $1 \to 0.25$. The prizes never change.

Most people pick **B** in Problem 1 and **A** in Problem 2. The ranking flips, although the only thing that changed is a common factor.

The Allais paradox – why it is a violation

Normalise $u(0) = 0$. The two modal choices say:

$$\begin{aligned} &\text{B in Problem 1:} && u(3000) > 0.80\,u(4000) \\ &\text{A in Problem 2:} && 0.20\,u(4000) > 0.25\,u(3000) \qquad \Big| \times 4 \\ & && 0.80\,u(4000) > u(3000) \end{aligned}$$

The first and the last line are the **same inequality with opposite signs**. No utility function can satisfy both.

Note what this rules out. The problem is *not* that people are too risk averse: u is arbitrary here, and it cancels out. Curvature of u cannot rescue expected utility, so the fix has to come from outside it, which is what Prospect Theory provides.

Exercise 3.1

Two decision problems

Problem 1: A. win €5,000 with probability 80% B. win €4,000 with certainty

Problem 2: A. win €5,000 with probability 16% B. win €4,000 with probability 20%

The agent has $u(x) = \sqrt{x \cdot 0.001}$.

Compute the expected utility of all four alternatives, and say what this agent chooses in each problem.

Exercise 3.1 – step by step

Step 1. Utilities of the payoffs, $u(x) = \sqrt{0.001\,x}$:

$$u(5000) = \sqrt{5} = 2.2361 \qquad u(4000) = \sqrt{4} = 2 \qquad u(0) = 0$$

Step 2. Problem 1: A: $0.80 \times 2.2361 + 0.20 \times 0 = 1.789$; B: $1 \times 2 = 2.000 \Rightarrow \mathbf{B}$.

Step 3. Problem 2: A: $0.16 \times 2.2361 + 0.84 \times 0 = 0.358$; B: $0.20 \times 2 + 0.80 \times 0 = 0.400 \Rightarrow \mathbf{B}$.

Step 4. Compare the two problems: every probability of Problem 1 is multiplied by 0.2 ($0.8 \to 0.16$, $1 \to 0.2$). So every expected utility is multiplied by 0.2: $1.789 \times 0.2 = 0.358$, $2 \times 0.2 = 0.4$.

Step 5. Multiplying both sides of an inequality by the same positive number keeps its direction: an expected-utility maximiser who picks B in Problem 1 picks B in Problem 2.

Typical mistake: forgetting the zero payoff; it contributes $u(0) = 0$ to every expected utility.

Exercise 3.2

What the data say

In the experiments of Kahneman and Tversky the modal respondent chose **B in Problem 1** and **A in Problem 2**.

(a) Is this pattern compatible with expected utility theory?

(b) Which axiom does it violate, and why?

Exercise 3.2 – step by step

(a) **Step 1.** Write the first choice as an inequality (using $u(0) = 0$): B in Problem 1 $\iff u(4000) > 0.8\,u(5000)$.

Step 2. Write the second choice: A in Problem 2 $\iff 0.16\,u(5000) > 0.20\,u(4000)$.

Step 3. Multiply the second by 5: $0.8\,u(5000) > u(4000)$.

Step 4. Steps 1 and 3 say opposite things about the same two numbers: no u can satisfy both $\Rightarrow$ **not compatible** with expected utility, whatever the utility function.

(b) **Step 1.** Rewrite Problem 2 as Problem 1 with a common component: with probability 0.8 you get 0 whatever you chose; with probability 0.2 you play Problem 1. Check: $0.2 \times 0.8 = 0.16$, $0.2 \times 1 = 0.2$.

Step 2. The **independence axiom** says that adding the same component to both options cannot change the ranking. The modal choices change it.

Prospect Theory – the two ingredients

Lecture 2 – “Prospect theory”; Lecture 3a – “probability misperception”, “reference dependence”

Probability weighting

$$w(p) = \frac{p^{\omega}}{[p^{\omega} + (1-p)^{\omega}]^{1/\omega}}$$

- $\omega \in (0, 1]$; $\omega = 1$ gives no curvature
- estimates: $\omega \approx 0.7$–0.8
- inverse-S: low p *over*-weighted, high p *under*-weighted

Reference dependence

- gains and losses from a reference point, not final wealth
- loss aversion: $|v(-x)| > v(x)$ for $x > 0$
- reflection effect: risk aversion over gains becomes risk seeking over losses

Exercise 3.3

Two Prospect Theory explanations

Explain, in two or three lines each, how Prospect Theory accounts for the pattern in Exercise 3.2:

(a) through probability misperception;

(b) through reference dependence.

Exercise 3.3 – step by step

(a) Probability misperception (use $w(p)$ with $\omega = 0.7$, as in 3.4)

Step 1. Problem 1, option B is certain: probability 1, nothing to distort.

Step 2. Option A has $p = 0.8$, a high probability: under-weighted, $w(0.8) = 0.68$. A is worth $0.68 \times 5000 \approx 3380 < 4000$: even a risk-neutral agent picks B.

Step 3. Problem 2 without distortion: $0.16 \times 5000 = 800 = 0.20 \times 4000$, a tie. With distortion both small probabilities are over-weighted, the smaller one proportionally more: $w(0.16) = 0.224$, $w(0.20) = 0.256$, so A is worth $0.224 \times 5000 \approx 1118$ against $0.256 \times 4000 \approx 1024 \Rightarrow$ A.

(b) Reference dependence

Step 1. Problem 1: the certain €4,000 becomes the reference point.

Step 2. Relative to it, the 0 of option A is a *loss* of €4,000; with loss aversion it weighs a lot $\Rightarrow$ B.

Step 3. Problem 2: no certain option, so no shift of the reference point; zero is not coded as a loss and A can be chosen.

Exercise 3.4

The weighting function

Let

$$w(p) = \frac{p^{\omega}}{[p^{\omega} + (1-p)^{\omega}]^{1/\omega}} \qquad \omega = 0.7$$

Compute $w(0.1)$ and $w(0.9)$, and say in each case whether the probability is over- or underweighted.

Exercise 3.4 – step by step

Step 1. Powers with $\omega = 0.7$: $0.1^{0.7} = 0.1995$, $0.9^{0.7} = 0.9289$.

Step 2. Denominator. It is the *same* for $p = 0.1$ and $p = 0.9$, because $1 - 0.1 = 0.9$:

$$\left[0.1^{0.7} + 0.9^{0.7}\right]^{1/0.7} = \left[1.1284\right]^{1.4286} = 1.1885$$

Step 3. $w(0.1) = 0.1995/1.1885 = 0.168$.

Step 4. $w(0.9) = 0.9289/1.1885 = 0.782$.

Step 5. Compare with the true probabilities: $0.168 > 0.1 \Rightarrow$ **overweighted**; $0.782 < 0.9 \Rightarrow$ **underweighted**.

Calculator tip: $x^{0.7}$ is `x ^ 0.7`; $1/0.7 = 1.4286$. Check: with $\omega = 1$ the formula gives back $w(p) = p$.

The fourfold pattern

	Gains	**Losses**
Low probability	risk seeking *buying a lottery ticket*	risk averse *buying insurance*
High probability	risk averse *settling a lawsuit you expect to win*	risk seeking *gambling to avoid a sure loss*

Probability weighting $\times$ diminishing sensitivity.

Exercise 3.5

The reflection effect

Reflect the two problems around zero:

Problem 3: A. $-4{,}000$ with prob. 80% B. $-3{,}000$ with certainty

Problem 4: A. $-4{,}000$ with prob. 20% B. $-3{,}000$ with prob. 25%

What did Kahneman and Tversky observe, what is the phenomenon called, and how does it differ from loss aversion?

Exercise 3.5 – step by step

Step 1. Notice the structure: Problems 3 and 4 are Problems 1 and 2 of the lecture with every payoff multiplied by -1 ("reflected" around zero).

Step 2. Recall the data. Gains: 80% chose the safe B in Problem 1. Losses: only 8% chose the safe B in Problem 3; in Problem 4, 58% chose B.

Step 3. Read it: risk aversion over gains becomes risk seeking over losses $\Rightarrow$ **reflection effect**.

Step 4. Distinguish from loss aversion: loss aversion compares a gain and a loss of the same size, $|v(-x)| > v(x)$. Here every outcome is a loss; the point is the attitude towards risk among losses.

Design caveat (likely oral question): the payoffs were hypothetical; real losses require an initial stake, which raises the house-money effect.

PART 4

Bayes' rule, base rates, representativeness

Bayes' rule

Lecture 3b – "The Bayes' rule", "A counting heuristic"

$$\underbrace{\Pr(A \mid c)}_{\text{posterior}} = \frac{\overbrace{\Pr(c \mid A)}^{\text{likelihood}} \overbrace{\Pr(A)}^{\text{prior}}}{\Pr(c \mid A)\Pr(A) + \Pr(c \mid B)\Pr(B)}$$

Independent draws $S = (c_1, \ldots, c_n)$ with replacement:

$$\Pr(S \mid A) = \prod_{t=1}^{n} \Pr(c_t \mid A)$$

Updating draw by draw (each posterior becomes the next prior) and updating on the whole sequence in one step give the **same** answer.

Two systematic mistakes

Lecture 3b – “Base-rate fallacy”; Lecture 4a – Grether (1980)

- **Base-rate fallacy**: $\Pr(A)$ is neglected, and the answer is given as if only the reliability of the signal mattered.
- **Representativeness**: $\Pr(A \mid c)$ is judged by how much c *resembles* what A typically produces.

Grether (1980), \$15 for a correct prediction and \$5 otherwise:

representativeness and Bayes **agree**	$\approx$ 80% correct
representativeness and Bayes **conflict**	$\approx$ 60% correct

Exercise 4.1

Base rates

A screening test for a rare condition is administered to the whole population.

- The condition affects 2 people in every 1,000.
- Anyone who has the condition tests positive: there are no false negatives.
- 5% of the people who do *not* have the condition also test positive.

A randomly selected person tests positive. What is the probability that this person has the condition?

Exercise 4.1 – step by step

Step 1. Translate the data into probabilities: $\Pr(D) = 0.002$, $\Pr(+ \mid D) = 1$, $\Pr(+ \mid \text{no } D) = 0.05$.

Step 2. Counting version, 10,000 people: with D: $10{,}000 \times 0.002 = 20$, all positive. Without D: $9{,}980$, of whom $9{,}980 \times 0.05 = 499$ positive.

Step 3. Total positives: $20 + 499 = 519$; genuine ones: 20. $\Pr(D \mid +) = 20/519 = 0.0385$.

Step 4. Same with Bayes' rule:

$$\Pr(D \mid +) = \frac{\Pr(+ \mid D)\Pr(D)}{\Pr(+ \mid D)\Pr(D) + \Pr(+ \mid \text{no } D)\Pr(\text{no } D)}$$

$$= \frac{1 \times 0.002}{1 \times 0.002 + 0.05 \times 0.998} = \frac{0.002}{0.0519} = 0.0385$$

Step 5. Less than 4%.

Typical mistake: answering 95% (one minus the false positive rate): that ignores the base rate, $\Pr(D) = 0.002$.

Exercise 4.2

Two cups – the setting

Two cups contain equally sized Amber (a) and Blue (b) marbles.

Cup A: two Ambers and one Blue Cup B: one Amber and two Blues

A die is thrown to select the cup: if the number drawn is 1 or 2 we use Cup A, otherwise Cup B. The selection is hidden. You then observe draws *with replacement* from the selected cup.

(a) What is the prior probability that Cup A is in use?
(b) The sequence observed is **amber – amber – blue**. Before computing anything: which cup *looks* more likely, and why?

Exercise 4.2 – step by step

(a) **Step 1.** A die has 6 equally likely faces; faces 1 and 2 select Cup A.

Step 2. $\Pr(A) = 2/6 = 1/3$, $\Pr(B) = 1 - 1/3 = 2/3$.

(b) **Step 1.** Composition of the sample: 2 amber out of $3 = 2/3$ amber.

Step 2. Composition of the cups: A is 2/3 amber, B is 1/3 amber. The sample "matches" Cup A.

Step 3. So Cup A *looks* more likely: the sample looks representative of A.

Step 4. What this ignores: the prior. Before any draw, B is twice as likely as A. Whether the sample is enough to overturn the prior is what Exercise 4.3 computes.

Exercise 4.3

Two cups – updating

Same setting as Exercise 4.2.

Use Bayes' rule to compute $\Pr(A \mid a, a, b)$.

Is Cup A or Cup B more likely to be in use?

Exercise 4.3 – step by step

Step 1. Likelihood of each draw: $\Pr(a \mid A) = \frac{2}{3}$, $\Pr(b \mid A) = \frac{1}{3}$, $\Pr(a \mid B) = \frac{1}{3}$, $\Pr(b \mid B) = \frac{2}{3}$.

Step 2. Likelihood of the sequence $S = (a, a, b)$, independent draws:

$$\Pr(S \mid A) = \tfrac{2}{3} \cdot \tfrac{2}{3} \cdot \tfrac{1}{3} = \tfrac{4}{27} \qquad \Pr(S \mid B) = \tfrac{1}{3} \cdot \tfrac{1}{3} \cdot \tfrac{2}{3} = \tfrac{2}{27}$$

Step 3. Multiply by the priors: A: $\frac{4}{27} \cdot \frac{1}{3} = \frac{4}{81}$; B: $\frac{2}{27} \cdot \frac{2}{3} = \frac{4}{81}$.

Step 4. Bayes' rule: $\Pr(A \mid S) = \dfrac{4/81}{4/81 + 4/81} = \dfrac{1}{2} = 0.5$.

Step 5. Interpret: the sample is twice as likely under A (4 against 2), the prior is twice as favourable to B (2/3 against 1/3): the two cancel exactly. Neither cup is more likely.

Representativeness (Exercise 4.2) says "Cup A": here the two criteria disagree, as in Grether's design.

Exercise 4.4

One more draw

A fourth draw is made from the same cup, and it is **amber**.

Update the probability that Cup A is in use.

Then check your answer a second way, by applying Bayes' rule to the full four-draw sequence in one step.

Exercise 4.4 – step by step

Sequential way

Step 1. The posterior after three draws is the new prior: $\Pr(A) = \Pr(B) = 0.5$.

Step 2. New draw: amber. $\Pr(a \mid A) = \frac{2}{3}$, $\Pr(a \mid B) = \frac{1}{3}$.

Step 3. $\Pr(A \mid a) = \dfrac{0.5 \times \frac{2}{3}}{0.5 \times \frac{2}{3} + 0.5 \times \frac{1}{3}} = \dfrac{1/3}{1/3 + 1/6} = \dfrac{1/3}{1/2} = \dfrac{2}{3}$.

One-step way, $S' = (a, a, b, a)$, original priors $\frac{1}{3}$ and $\frac{2}{3}$

Step 1. $\Pr(S' \mid A) = (\frac{2}{3})^3(\frac{1}{3}) = \frac{8}{81}$; $\Pr(S' \mid B) = (\frac{1}{3})^3(\frac{2}{3}) = \frac{2}{81}$.

Step 2. Times the priors: A: $\frac{8}{81} \cdot \frac{1}{3} = \frac{8}{243}$; B: $\frac{2}{81} \cdot \frac{2}{3} = \frac{4}{243}$.

Step 3. $\Pr(A \mid S') = \dfrac{8}{8+4} = \dfrac{2}{3}$. Same answer.

Exercise 4.5

Naming the bias

The tendency to make mistakes in the assessment of probability, driven by a rough sense of "similarity" between the observed signal and the underlying event, is called:

a. Loss aversion
b. Representativeness bias
c. Risk aversion
d. Probability matching

Complete the sentence: *in the setting of Exercises 4.2 and 4.3, this tendency would induce one to mistakenly attribute _____, because the observed sample of three draws has the same proportions as _____, and in this sense the sample looks _____.*

Exercise 4.5 – step by step

Step 1. Key words in the question: “similarity” between the signal and the event ⇒ **b. representativeness bias**.

Step 2. Discard the others: loss and risk aversion are about attitudes towards outcomes, not about assessing probabilities; probability matching is about repeated predictions.

Step 3. Fill the sentence using Exercises 4.2 and 4.3:

- what representativeness suggests: Cup A is more likely ⇒ “a higher probability to Cup A than to Cup B”
- why: 2 amber out of 3, the composition of **Cup A**
- in this sense the sample looks “**representative of Cup A**”

Step 4. The word “mistakenly”: by Bayes’ rule the two cups are equally likely (Exercise 4.3), so attributing a higher probability to A is an error.

Exam mechanics: round button = exactly one answer; square box = at least one. Wrong answers cost 50% of the points, blanks cost nothing.

PART 5

Probability matching

Probability matching

Lecture 4b – “Probability matching”, “Rational behavior”, “Siegel and Goldstein’s Experiment”

One event occurs with probability $p > 1/2$, the other with $1 - p$; $U_C > U_I$.

predict the more likely event	$pU_C + (1-p)U_I$
predict the less likely one	$pU_I + (1-p)U_C$
difference	$(2p-1)(U_C - U_I) > 0$

Always predict the more likely event, whatever the risk attitude: the two payoffs are the same in both lotteries.

Matching: predict each event with a frequency equal to its frequency of occurrence. Hit rate $p^2 + (1-p)^2 < p$.

Siegel and Goldstein: no-pay / pay / pay-loss. Stronger incentives move behaviour away from matching. Animals are always run with incentives, which is why rats look smarter than humans.

Exercise 5.1

Earnings under the two strategies

In a class experiment on probability matching each correct prediction pays €0.25, and teams make predictions over 20 trials. The more likely event was programmed to occur with probability 0.80.

(a) How much would a team earn *per trial* by being perfectly rational, once it has learnt which event is more likely?

(b) And a team that follows perfect probability matching?

Exercise 5.1 – step by step

(a) **Step 1.** Rational strategy: always predict the more likely event.

Step 2. It occurs with probability 0.80, so the prediction is correct 80% of the time: $0.80 \times €0.25 = €0.20$ per trial.

(b) **Step 1.** Matching: predict the likely event (L) 80% of the time, the other (R) 20%, independently of what happens.

Step 2. List the four cases:

prediction	event	probability	correct?
L	L	$0.8 \times 0.8 = 0.64$	yes
L	R	$0.8 \times 0.2 = 0.16$	no
R	L	$0.2 \times 0.8 = 0.16$	no
R	R	$0.2 \times 0.2 = 0.04$	yes

Step 3. Probability of a correct prediction: $0.64 + 0.04 = 0.68 \Rightarrow 0.68 \times €0.25 = €0.17$ per trial.

Step 4. Over 20 trials: $20 \times 0.20 = €4.00$ against $20 \times 0.17 = €3.40$.

Exercise 5.2

Generalise

Keep the payoff at €0.25 and let p denote the probability of the more likely event.

Write the per-trial expected earnings under the two strategies, and show that matching is always worse for $p > 1/2$.

At which p is the loss largest?

Exercise 5.2 – step by step

Step 1. Rational: correct with probability $p \Rightarrow p \times 0.25$.

Step 2. Matching (as in 5.1): correct with probability $p \cdot p + (1-p)(1-p) \Rightarrow \left[p^2 + (1-p)^2\right] \times 0.25$.

Step 3. Difference of the probabilities:

$$p - \left[p^2 + (1-p)^2\right] = p - p^2 - (1 - 2p + p^2) = -2p^2 + 3p - 1$$

Step 4. Factorise: $-2p^2 + 3p - 1 = (1-p)(2p-1)$. Check at $p = 0.8$: $0.2 \times 0.6 = 0.12 = 0.80 - 0.68$.

Step 5. Sign: for $1/2 < p < 1$, both $(1-p) > 0$ and $(2p-1) > 0 \Rightarrow$ the difference is positive: matching is always worse.

Step 6. Largest loss: derivative $-4p + 3 = 0 \Rightarrow p = 3/4$; the difference is $(1/4)(1/2) = 0.125$, i.e. $0.125 \times 0.25 = €0.031$ per trial.

Check with the lecture: $p = 0.75$, €0.20 per correct answer: rational $0.75 \times 0.20 = 0.15$, matching $0.625 \times 0.20 = 0.125$.

PART 6

Oral questions

Sample oral questions I

The Allais paradox

1. Illustrate what the Allais paradox is.
2. What is the independence axiom?
3. Explain why the Allais paradox represents a violation of the independence axiom.
4. Illustrate how reference dependence, as assumed by Prospect Theory, can provide an explanation of the Allais paradox.

Risk attitudes and the lottery-choice menu

1. How does expected utility theory represent risk aversion?
2. Describe the Holt and Laury lottery-choice task. What does the switching point tell us?
3. What did Holt and Laury find when they scaled payoffs up, and what changes when the payoffs are hypothetical?

Sample oral questions II

Information and learning

1. What is the base-rate fallacy? Give an example.
2. What is representativeness bias, and how did Grether design an experiment that separates it from Bayesian updating?

Probability matching

1. What is the behavioural bias that goes under the name of “probability matching”?
2. Can you describe the experiment used to study this phenomenon?
3. What is the rational behaviour in this experiment, and what behaviour was observed instead among many respondents?
4. Simple animals like chicks and rats did not fall prey to probability matching, while humans did. Which explanation was given for this surprising result?